



Educational Platform for Education Providers

Data Flow Diagrams (DFD)

System architecture, data flows, processes, and data stores across the eSlate platform — including the React frontend, Express.js backend (PostgreSQL), and Laravel API backend (SQLite).

6

ACTORS

15

PROCESSES

2

DATABASES

37+

TABLES

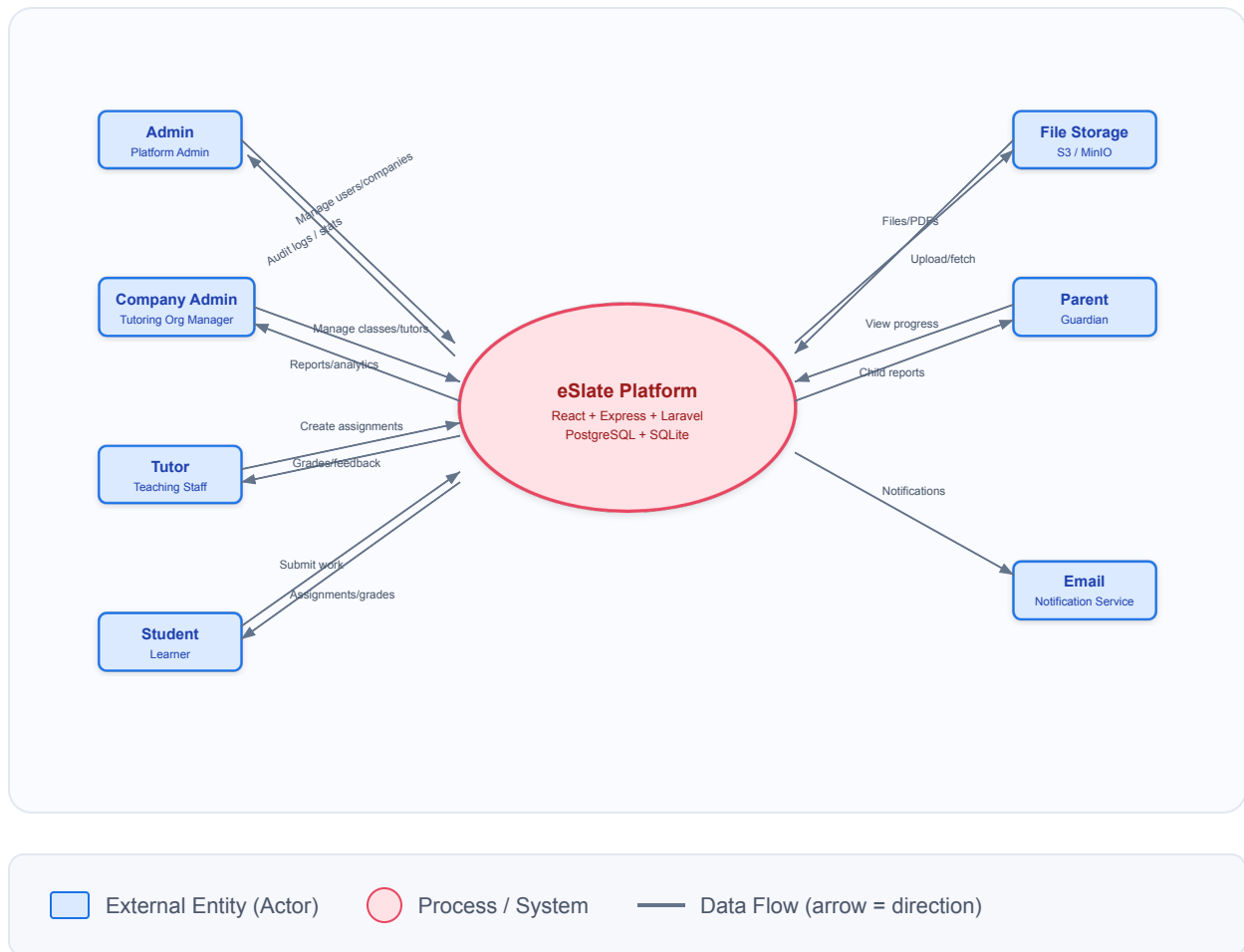
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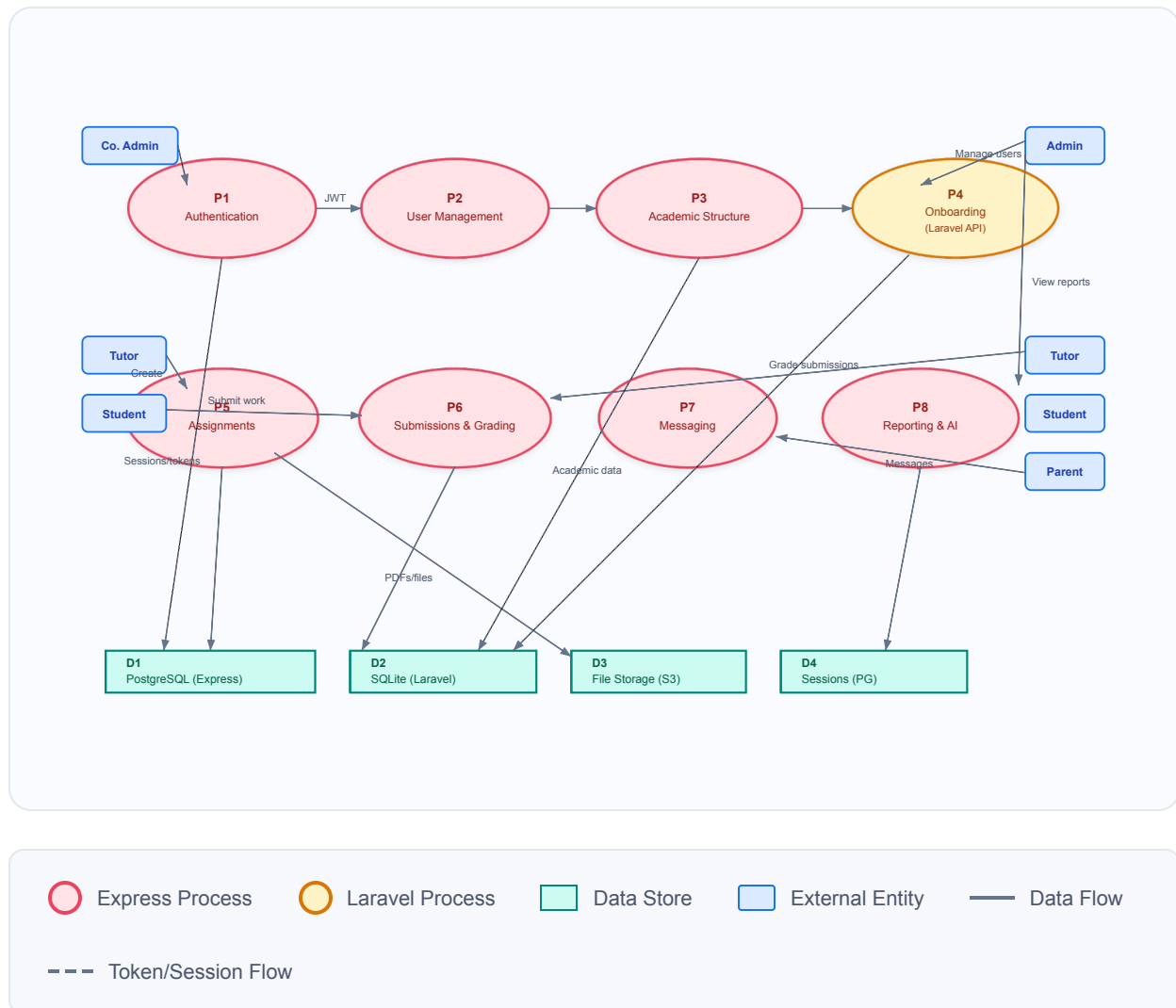
1 · Level 0 — Context Diagram

The entire eSlate system as a single process showing all external entities and top-level data flows



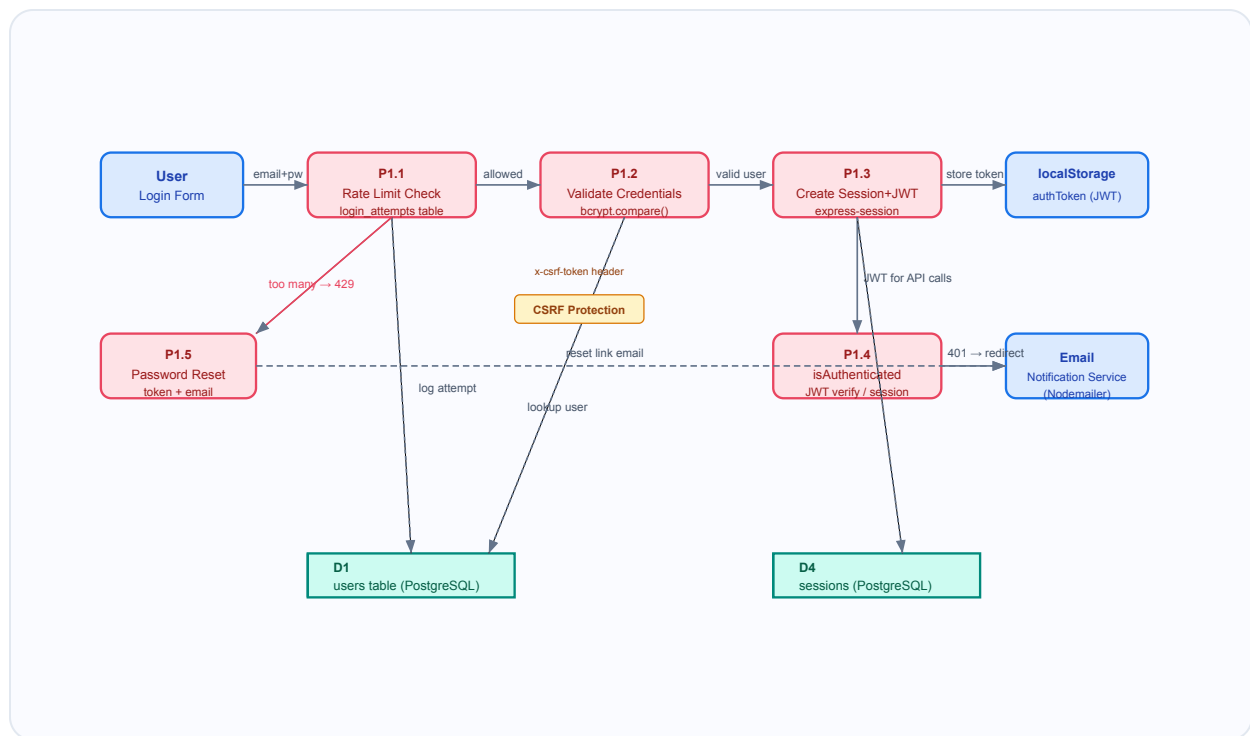
2 · Level 1 — System Overview

eSlate decomposed into its primary process groups showing flows between actors, processes, and data stores



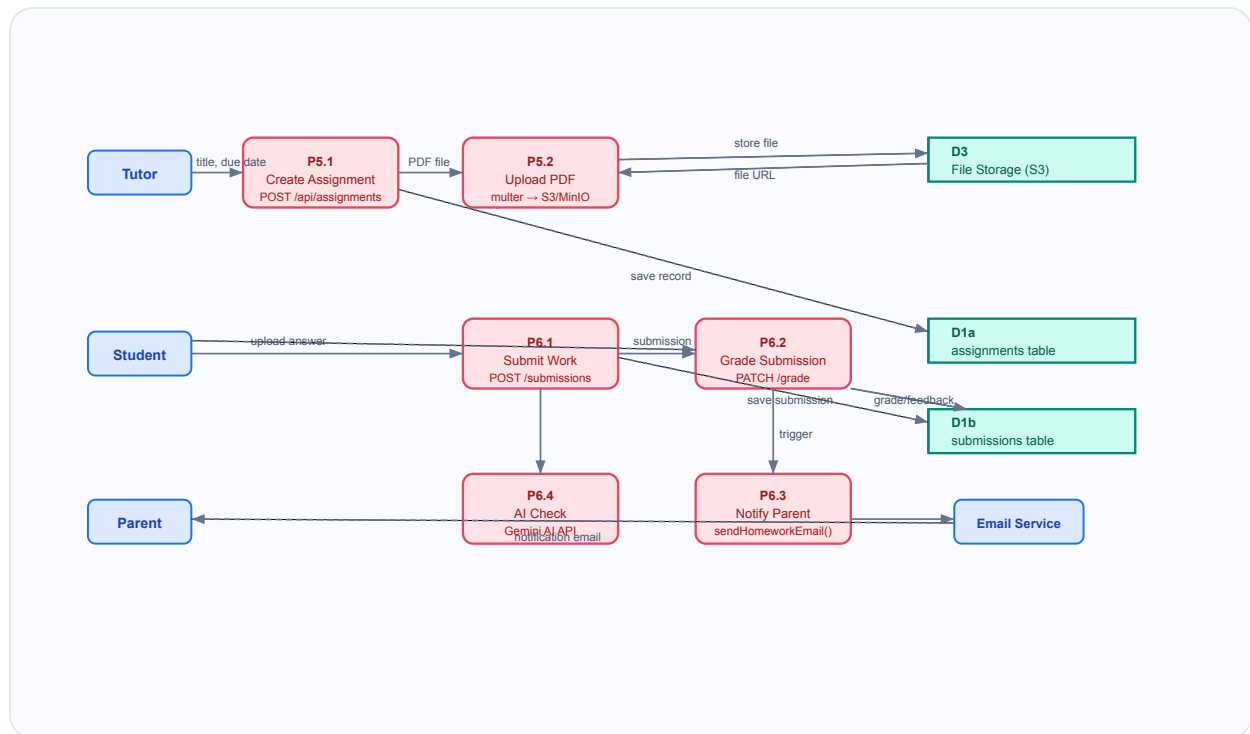
3 · Level 2A — Authentication & Session Flow

Detailed data flows within P1: Authentication — login, JWT, session, rate-limiting, password reset



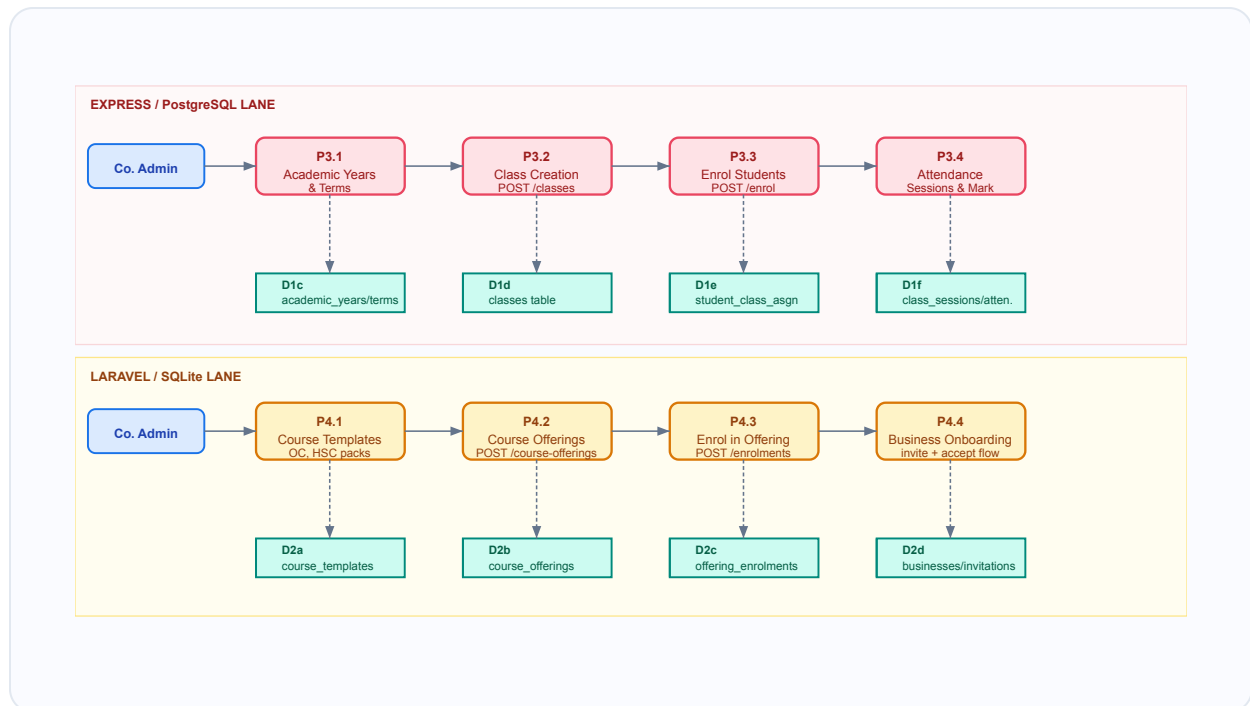
4 · Level 2B — Assignment & Submission Flow

Detailed flows within P5 (Assignments) and P6 (Submissions & Grading) including file handling and notifications



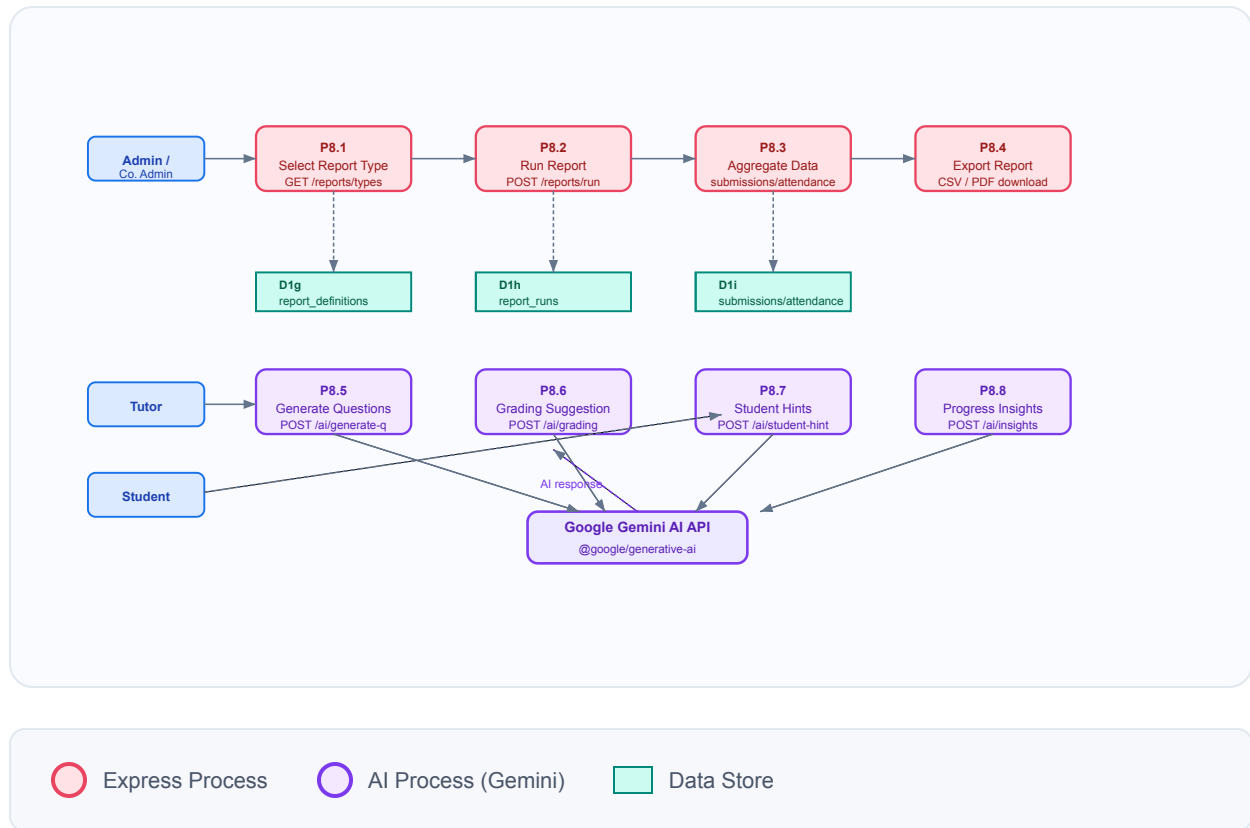
5 · Level 2C — Academic Structure & Class Management

Data flows for academic years, terms, classes, enrolments and the Laravel course/offering system



6 · Level 2D — Reporting & AI Features

Data flows within P8: report generation pipeline and AI-assisted features (question generation, grading suggestions, hints)



7 · Data Dictionary

Key entities, processes, and data stores referenced in the DFDs above

External Entities

Entity	Role	Interacts With
ADMIN	Platform-level administrator (admin@eslate.com)	P1, P2, P8 — user/company management, audit logs, stats
COMPANY ADMIN	Manages a single tutoring organisation	P2, P3, P4, P8 — staff, classes, reports
TUTOR	Teaching staff; creates assignments, grades work	P5, P6, P7, P8.5/P8.6 — assignments, grading, AI
STUDENT	Learner; submits work, takes tests	P5, P6, P8.7 — submissions, hints
PARENT	Guardian; monitors progress, receives notifications	P6.3, P7 — grades, messaging
EMAIL SERVICE	Nodemailer; sends notifications, reset links	P1.5, P6.3 — password reset, submission alerts

Key Data Stores

ID	Store	Technology	Key Tables
D1	Express PostgreSQL	PostgreSQL + Drizzle ORM	users, assignments, submissions, classes, messages, reports
D2	Laravel SQLite	SQLite + Eloquent ORM	businesses, academic_years, course_templates, offering_enrolments
D3	File Storage	S3 / MinIO	PDFs, annotated submissions, exported reports
D4	Session Store	PostgreSQL (connect-pg-simple)	sessions (sid, sess JSON, expire)

Primary Data Flows

Flow	From → To	Data Carried	Protocol
Login	User → P1.2 → D1	email, password (bcrypt), JWT token	HTTPS JSON
Session Auth	Browser → P1.4 → D4	session cookie / Bearer JWT	HTTP Cookie / Header
Assignment Create	Tutor → P5.1 → D1 + D3	title, due date, PDF file (multipart)	HTTPS multipart
Submission	Student → P6.1 → D1 + D3	answer file, annotations (JSON)	HTTPS JSON/multipart
Grading	Tutor → P6.2 → D1 → P6.3 → Email	score, feedback text, notif. trigger	HTTPS JSON + SMTP
Report Run	Admin → P8.2 → D1 → P8.4 → Browser	report params, aggregated CSV/JSON	HTTPS JSON + file download

AI Request	Tutor/Student → P8.5-8 → Gemini	prompt, submission text, rubric	HTTPS (Google API)
Laravel Auth	User → POST /api/login → D2	email, password → Sanctum token	HTTPS JSON